

AI DM Conversation Agent — Backend Technical Brief

Project Goal

Build an AI-powered DM / conversation agent capable of answering questions, retrieving knowledge, and routing users to human agents when needed.

Backend Stack

Runtime: Python 3.11
Framework: FastAPI
Server: Uvicorn
Environment: python-dotenv
AI Access: OpenAI API

Architecture (MVP)

Simple API service. No queues, workers, or microservices in the first phase.

Core Endpoint

POST /chat → receives message → sends to AI service → returns response.

Minimal Project Structure

project-root
main.py
routes/chat.py
services/ai_agent.py
.env
requirements.txt

Request Flow

Client → POST /chat → Route → Controller → AI Service → Response

Deployment Target

Initial deployment compatible with standard hosting environments (e.g., cPanel Python app).

Development Principles

Start minimal, avoid premature architecture, reach working MVP quickly.